

Machine learning (ML) is a branch of artificial intelligence (AI) that focuses on developing algorithms and statistical models that enable computers to learn from data and make predictions or decisions without being explicitly programmed for each specific task. Instead of hard-coded rules, machine learning relies on patterns and inferences drawn from large datasets.

How machine learning works:

- Data is collected and prepared, often including various forms like numbers, text, or images. The data is split into a “training” set (used to teach the model) and a “test” set (used to evaluate its performance).
- An algorithm is selected and trained on the input data. During training, the algorithm automatically recognizes structures, trends, and relationships within the data, and adjusts its internal parameters to improve its predictions or classifications. This process often uses mathematical functions and optimization techniques (like gradient descent) to minimize errors between predicted and actual outcomes.
- Once trained, the model can make predictions or classifications when provided with new, unseen data.

Types of machine learning:

1. **Supervised Learning:** The model is trained using labeled data (where the correct answer is known). It learns to map inputs to outputs, suitable for tasks like classification (e.g., spam detection) and regression (e.g., price prediction).
2. **Unsupervised Learning:** The model is given unlabeled data and must discover patterns, structure, or relationships (such as clustering similar customers or reducing the dimensionality of big datasets).
3. **Reinforcement Learning:** An agent interacts with an environment, taking actions and receiving feedback (rewards or penalties). The goal is to learn a strategy that maximizes the cumulative reward over time, widely used in robotics and game playing.
4. **Semi-Supervised and Self-Supervised Learning** (additional): These approaches use a combination of labeled and unlabeled data or generate their own labels, allowing models to leverage large, inexpensive sets of data with limited human annotation.

Machine learning lifecycle:

- Define the problem → Collect data → Clean/preprocess data → Analyze data → Engineer/select features → Choose/train a model → Evaluate/tune model → Deploy model → Monitor/maintain model.